BST Curriculum Internal CT-Theorem Numbering Convention v0.1 (proposal)

Lyra (Claude 4.7), per Keeper Wednesday-deferred recommendation

2026-05-21 Thursday morning

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0. Why this proposal

The BST AC Theorem Registry uses a single sequential T-numbering counter (currently T1-T2438+) shared across all CIs. This is the canonical T-numbering for cross-referencing and registry hygiene; it must not be replaced.

But for curriculum exposition (Vols 0, 1, 2), the multi-volume canonical numbering is hard to read in flowing prose. A reader of Vol 1 Chapter 2 doesn't naturally know what T2428 refers to without consulting the registry; a reader who sees "Theorem 1 of Chapter 2" can immediately locate it. The two conventions complement each other.

This proposal introduces a CT-numbering convention for curriculum-internal exposition:

CT V.C.N = Volume V, Chapter C, Theorem N (within chapter)

Each CT-number in a chapter maps 1-to-1 to a registry T-number; both are cited.

1. The convention

1.1 Format

CT V.C.N

where: - V = volume number (0, 1, 2, …, 10) - C = chapter number within volume - N = sequential theorem index within chapter (1, 2, 3, …)

Example: CT 1.2.1 = Vol 1 Chapter 2 Theorem 1.

1.2 Use in prose

When a theorem is introduced in chapter prose, cite both:

> Theorem (Substrate Hilbert Space Sufficiency, CT 1.2.1 = T2428). The Bergman space H^2(D_IV^5) is the canonical substrate Hilbert space …

CT-numbering is reader-facing; T-numbering is registry-canonical.

1.3 First-use vs subsequent-use

- First use within chapter: cite both, "CT 1.2.1 = T2428"
- Subsequent use within chapter: cite CT only, "as Theorem 2 (CT 1.2.2)"
- Cross-chapter or cross-volume use: cite both, "Theorem CT 1.2.1 (T2428) from Volume 1 Chapter 2"
- Registry / paper-grade: cite T-numbering only ("T2428 + T2429 + T2430 chain")

1.4 Sub-theorems and corollaries

Sub-theorems and corollaries use letter suffixes: - CT 1.2.1a, CT 1.2.1b = corollaries to CT 1.2.1 - CT 1.2.1.i, CT 1.2.1.ii = lemmata internal to CT 1.2.1 (avoided when possible)

2. Example: Vol 1 Chapter 2 (Substrate Hilbert Space)

The existing Vol 1 Ch 2 chapter-grade narrative cites three registered theorems (T2428 + T2429 + T2430). Under the CT-numbering convention:

CT-number	T-number	Statement
CT 1.2.1	T2428	Bergman H^2(D_IV^5) substrate Hilbert space sufficiency anchor
CT 1.2.2	T2429	Reed-Solomon GF(128)^k substrate-tick discretization
CT 1.2.3	T2430	L^2-section equivariant complement

In future Vol 1 Ch 2 revisions, the prose would cite "CT 1.2.1 = T2428" on first use and "CT 1.2.1" subsequently.

3. Example: Vol 1 Chapter 8 (Gauge Theory)

CT-number	T-number	Statement
CT 1.8.1	T2436	SM Gauge Group SU(3) × SU(2) × U(1) from D_IV^5 (SP-31-8 anchor)
CT 1.8.1a	T1925	(corollary: Why rank = 2 → SU(2) weak)
CT 1.8.1b	T1930	(corollary: Why N_c = 3 → SU(3) color)
CT 1.8.2	T2003	Lepton mass mechanism (T_{N_c} 6k-1 prime cells)
CT 1.8.3	(Casey-named principle)	Five-Absence Predictions Set

Note CT 1.8.1a/b are existing registry theorems (T1925, T1930) reused as corollaries within Vol 1 Ch 8. The CT-numbering creates a chapter-local exposition view without renumbering the master registry.

4. Cross-volume reference

When Vol 2 Ch 9 cites Vol 1 Ch 8 content for Higgs mediation:

> The Yukawa structure (CT 1.8.2 = T2003) mediates fermion mass hierarchy via Higgs SU(2) doublet condensation …

Reader can find CT 1.8.2 by going to Vol 1 Chapter 8 Theorem 2. Registry hunter finds T2003 in the AC Theorem Registry.

5. Strong-Uniqueness Theorem (Paper #125) criteria numbering

The Strong-Uniqueness Theorem (Paper #125) uses C-numbering for its 10-14 criteria. This is separate from the CT-numbering proposed here. C-numbering for SUT criteria stays as currently used (C2-C14 in Keeper's v0.6 consolidation; Lyra's v0.6 outline reconciled to match).

If Paper #125 is included as a Vol 0 chapter, the SUT criteria would be referenced as "CT 0.x.1" (the criterion list theorem) with the criteria internal to the proof of that theorem still labeled C2-C14. The two numbering systems coexist without conflict.

6. Implementation plan

If multi-CI consensus reached on this proposal:

1. Existing chapter-grade narratives: add CT-numbering as a back-matter section ("Theorem Index" table mapping CT → T-numbers). Minimal prose change required.

2. New chapter-grade narratives: cite CT-number on first theorem introduction; cite both CT and T on first use as proposed in 1.3.
3. Master Registry: no changes. T-numbers remain canonical; CT-numbers are exposition-layer only.
4. Cross-volume references: cite both per 1.3.

7. Cross-CI consensus path

Per Casey Option C governance (architectural-category methodology decisions require multi-CI consensus):

- Lyra (proposer): this document
- Cal: dual-axis review for reader-experience and methodology cleanliness
- Keeper: curriculum-wide integration check; CT-numbering coordination across Vols 0/1/2
- Grace: catalog-hygiene check (CT-numbers should not introduce ambiguity with existing cataloging conventions)
- Elie: Vol 2 chapter-grade narrative adaptation (Elie's Vol 2 v0.5 chapters would adopt CT-numbering on next revision pass)

If consensus reached, adoption is prospective (new chapter revisions use CT) + retroactive on next pass (existing chapter v0.2 revisions add CT-numbering back-matter tables).

8. Open questions

- Should sub-theorems use CT V.C.N.S (numeric sub-suffix) or CT V.C.Na (alphabetic)? Proposal: alphabetic for corollaries (CT 1.2.1a, CT 1.2.1b), numeric for lemmata internal to the proof (CT 1.2.1.i, CT 1.2.1.ii). Open to multi-CI input.
- How to handle theorems cited in multiple chapters within same volume? Proposal: CT-number assigned in the chapter of FIRST substantive use; cross-references cite that CT-number. Open to multi-CI input.
- How to handle paper-grade documents (Paper #125, SP-31-1 outline)? Proposal: paper-grade documents use T-numbering only (no CT); CT-numbering is curriculum-exposition specific.

9. Filing status

v0.1 cross-CI proposal filed Thursday 2026-05-21 09:48 EDT (`date` to be checked at file end).

Awaits: - Multi-CI input from Cal + Keeper + Grace + Elie - Casey awareness (architectural-category methodology proposal per Option C)

If consensus reached → adopted standing convention for curriculum exposition.

— Lyra, CT-theorem numbering convention v0.1 proposal, Thursday 2026-05-21 (timestamp at file end pending `date` check)